

SCENARIO BASED ASSESSMENT

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1. A hospital wants to estimate consultation charges for different patient categories. Design a C++ program for a Consultation Charge System. Design a C++ program for hospital management

Task:

- A. Use function overloading to calculate charge for general consultation and specialist
- B. consultation.
- C. Use call by reference to modify the final bill amount.
- D. Use default arguments to add a standard service charge if needed.
- E. Use function prototyping for all calculation functions.
- F. Display the patient type and total consultation charge.(BL3)

code:

PROGRAM EDITOR

C++

Run

Save

INPUT

```
1 #include <iostream>
2 using namespace std;
3 int charge(int);
4 int charge(int, int);
5 void bill(int &);
6 int service(int, int = 100);
7 int charge(int x) { return x; }
8 int charge(int x, int y) { return x + y; }
9 void bill(int &b) { b += 50; }
10 int service(int b, int s) { return b + s; }
11 int main() {
12     int ch, total;
13     cin >> ch;
14     if (ch == 1) {
15         int fee;
16         cin >> fee;
17         total = charge(fee);
18         cout << "General\n";
```

1
500

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```

19     } else {
20         int fee, sp;
21         cin >> fee >> sp;
22         total = charge(fee, sp);
23         cout << "Specialist\n";
24     }
25     bill(total);
26     total = service(total);
27     cout << total;
28     return 0;
29 }

```

OUTPUT

```

General
650

```

2. Design a C++ program Fitness Centre Membership System. A gym wants to manage member subscriptions, track plans, and calculate membership fees.

Task:

- Create a class Member with attributes: Member ID, Name, Plan Type, and Fee.
- Implement member functions to assign plans, calculate fees, and display membership details. Use an array of objects to manage multiple members.
- Add validation for valid membership plans.
- Include a static member to track total active members.
- Evaluate how your design supports easy modification of pricing plans.

Code:

PROGRAM EDITOR

C++

Run

Save

```

1 #include <iostream>
2 using namespace std;
3 class Member {
4     int id;
5     string name, plan;
6     float fee;
7 public:
8     static int count;
9     void get() {
10         cin >> id >> name >> plan;
11         if (plan == "Basic") fee = 500;
12         else if (plan == "Premium") fee = 1000;
13         else if (plan == "Gold") fee = 1500;
14         else {
15             cout << "Invalid Plan\n";
16             fee = 0;
17         }
18         count++;
19     }

```

INPUT

```

2
101 Ram Basic
102 Ravi Premium

```

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```

20 void display() {
21     cout << id << " " << name << " " << plan << " " << fee << endl;
22 }
23 };
24 int Member::count = 0;
25 int main() {
26     int n;
27     cin >> n;
28     Member m[10];
29     for (int i = 0; i < n; i++)
30         m[i].get();
31     for (int i = 0; i < n; i++)
32         m[i].display();
33     cout << "Total Active Members = " << Member::count;
34     return 0;
35 }

```

OUTPUT

```

101 Ram Basic 500
102 Ravi Premium 1000
Total Active Members = 2

```

3. A bank wants to calculate loan repayment details for customers. Design a C++ program for a Loan Repayment System.

Task:

- Use function prototyping to declare EMI-related functions.
- Use default arguments for default interest rate where applicable.
- Apply call by reference to update the payable amount after subsidy.
- Use inline functions for simple interest calculations.
- Display the loan amount, interest, and repayment details clearly.

Code:

PROGRAM EDITOR

C++

Run

Save

```

1 #include <iostream>
2 using namespace std;
3 inline float interest(float, float = 8);
4 void subsidy(float &);
5 void display(float, float);
6 inline float interest(float loan, float rate) {
7     return loan * rate / 100;
8 }
9 void subsidy(float &amt) {
10     amt -= 1000;
11 }
12 void display(float loan, float total) {
13     cout << "Loan = " << loan << endl;
14     cout << "Repayment = " << total;
15 }
16 int main() {
17     float loan, payable;
18     cin >> loan;
19     payable = loan + interest(loan);

```

INPUT

50000

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```
20     subsidy(payable);
21     display(loan, payable);
22     return 0;
23 }
```

OUTPUT

```
Loan = 50000
Repayment = 53000
```

4. A delivery company wants to automate the calculation of parcel delivery charges based on parcel weight, delivery distance, and delivery type. Design and implement a C++ program using functions and object-oriented programming concepts to develop a Parcel Charge Management System.

Task:

- A. Create a class Parcel with appropriate private data members such as parcel type, weight, distance, and packaging charge.
- B. Use public member functions to read parcel details, calculate the delivery charge, and display the parcel information.
- C. Use function overloading to calculate charges for Normal Delivery and Express Delivery.
- D. Use default arguments to assign a default packaging charge when it is not specified by the user.
- E. Use an inline function to calculate the base charge based on parcel weight and delivery distance.
- F. Maintain the total number of parcels processed using a static data member and display it through a static member function.

Code:

```
PROGRAM EDITOR C++ Run Save INPUT
1 #include <iostream>
2 using namespace std;
3 class Parcel {
4     string type;
5     float weight, distance, pack;
6 public:
7     static int count;
8     void get(float p = 50) {
9         cin >> type >> weight >> distance;
10        pack = p;
11        count++;
12    }
13    inline float base() {
14        return weight * 10 + distance * 2;
15    }
16    float charge() {
17        return base() + pack;
18    }
19    float charge(int ex) {
20        return base() + pack + ex;
21    }
22    void display(float total) {
23        cout << "Parcel: " << type << endl;
24        cout << "Charge = " << total << endl;
25    }
26    static void totalParcels() {
27        cout << "Total Parcels = " << count;
28    }
29 };
30 int Parcel::count = 0;
31 int main() {
32     Parcel p;
33     int ch;
34     p.get();
35     cin >> ch;
36     if (ch == 1)
37         p.display(p.charge());
38 }
```

Box 5 z
2

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```

34     p.get();
35     cin >> ch;
36     if (ch == 1)
37         p.display(p.charge());
38     else
39         p.display(p.charge(100));
40     Parcel::totalParcels();
41     return 0;
42 }

```

OUTPUT

```

Parcel: Box
Charge = 240
Total Parcels = 1

```

5. A utility company wants to automate the calculation of water bills for domestic and commercial consumers. Design and implement a C++ program using functions and object-oriented programming concepts to develop a Water Billing System.

Task:

- create a class Consumer with appropriate private data members such as consumer ID, consumer type, water consumption (in liters), and bill amount.
- Use public member functions to input consumer details, calculate the water bill, and display the billing information.
- Implement function overloading to calculate water bills for Domestic and Commercial consumers using different tariff rates.
- Use call by reference to update the bill amount after applying for a government subsidy.
- Use default arguments to include a service tax, where a default tax rate is applied unless specified.
- Declare all functions using function prototyping before their definitions.
- Maintain the total number of consumers processed using a static data member and display it through a static member function.
- Create an array of objects to store and process the details of multiple consumers, and display the consumer type, water consumption, subsidy-adjusted bill amount.

Code:

```
1 #include <iostream>
2 using namespace std;
3 class Consumer {
4     int id;
5     string type;
6     float water, bill;
7 public:
8     static int count;
9     void get();
10    float calc(float);
11    float calc(float, int);
12    void subsidy(float &);
13    float tax(float, float = 5);
14    void display();
15    static void total();
16 };
17 int Consumer::count = 0;
18 void Consumer::get() {
19     cin >> id >> type >> water;
20     count++;
```

```
20     count++;
21     if (type == "Domestic")
22         bill = calc(water);
23     else
24         bill = calc(water, 1);
25     subsidy(bill);
26     bill = tax(bill);
27 }
28 float Consumer::calc(float w) { return w * 0.5; }
29 float Consumer::calc(float w, int) { return w * 1.0; }
30 void Consumer::subsidy(float &b) { b -= 50; }
31 float Consumer::tax(float b, float t) { return b + (b * t / 100); }
32 void Consumer::display() {
33     cout << type << " " << water << "L " << bill << endl;
34 }
35 void Consumer::total() {
36     cout << "Total Consumers = " << count;
37 }
38 int main() {
39     int n;
40     cin >> n;
```

```
2
101 Domestic 500
102 Commercial 500
```

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```
2
101 Domestic 500
102 Commercial 500
```

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```
40     cin >> n;
41     Consumer c[10];
42     for (int i = 0; i < n; i++)
43         c[i].get();
44     for (int i = 0; i < n; i++)
45         c[i].display();
46     Consumer::total();
47     return 0;
48 }
```

OUTPUT

```
Domestic 500L 210
Commercial 500L 472.5
Total Consumers = 2
```